The Lighthouse Lantern Mystery

Whodunnit? · Right Triangles & Trig · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Right Triangles & Trig	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Pythagorean theorem, special right triangles, sine, cosine and tangent ratios.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Dr. Gale took it, in the Cliff Path, using this method: Carried in a backpack.

The reveal (read aloud): In the fog, **Dr. Gale** slipped along the **Cliff Path**, packed the Great Lens carefully into a **backpack**, and hiked away before the tide came in. Every alibi and method you cleared led to him. A bright detective, indeed!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Keeper Odell · Place: Lamp Room
2	Method: Rowed away in a dinghy
3	Suspect: Captain Marlow · Place: Harbor Dock
4	Method: Hidden in a fish crate
5	Suspect: Ms. Brine · Place: Fish Market
6	Method: Lowered by rope
7	Suspect: Nico Reyes · Place: Boathouse
8	Method: Wrapped in a fishing net
9	Suspect: Mr. Tern · Place: Spiral Stairs
10	Method: Floated in a barrel

Answer Key

The Lighthouse Lantern Mystery · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	A right triangle has legs 6 and 8. Find the hypotenuse.	**D.** 10	$6^2 + 8^2 = 36 + 64 = 100$ 10	Keeper Odell was polishing the brass in the Lamp Room. The visitor logbook shows his time.
2	A right triangle has a leg of 5 and a hypotenuse of 13. Find the other leg.	**D.** 12	$5^2 + b^2 = 13^2$ $b^2 = 144$ 12	The dinghy is still tied up with the tide line untouched. No one rowed.
3	An isosceles right triangle has legs of length 5. Find the hypotenuse.	**A.** $5\sqrt{2}$	45°-45°-90°: hypotenuse = leg · $\sqrt{2}$ $5\sqrt{2}$	Captain Marlow was loading passengers at the Harbor Dock. All twenty remember him.
4	In a 45°-45°-90° triangle the hypotenuse is 8. Find the length of a leg.	**B.** $4\sqrt{2}$	leg = $\frac{\text{hypotenuse}}{\sqrt{2}}$ = $\frac{8}{\sqrt{2}}$ rationalize: $\frac{8\sqrt{2}}{2}$ $4\sqrt{2}$	Every fish crate was opened at the market. Only fish.
5	In a 30°-60°-90° triangle the shortest side is 4. Find the longer leg.	**A.** $4\sqrt{3}$	longer leg = short leg · $\sqrt{3}$ $4\sqrt{3}$	Ms. Brine was selling cod at the Fish Market. The cash register shows each sale.
6	In right triangle ABC with the right angle at C, AB = 5, BC = 3 and AC = 4. Find sin A.	**B.** $\frac{3}{5}$	sin A = $\frac{\text{opposite}}{\text{hypotenuse}}$ = $\frac{BC}{AB}$ $\frac{3}{5}$	The rope is coiled and dry. Nothing was lowered.
7	In right triangle ABC with the right angle at C, AB = 5, BC = 3 and AC = 4. Find tan A.	**C.** $\frac{3}{4}$	tan A = $\frac{\text{opposite}}{\text{adjacent}}$ = $\frac{BC}{AC}$ $\frac{3}{4}$	Nico Reyes was sanding a hull in the Boathouse with two apprentices.
8	A 10-ft ladder leans against a wall and makes a 60° angle with the ground. How high up the wall does it reach?	**C.** $5\sqrt{3}$ ft	$\sin 60° = \frac{h}{10}$ $h = 10 \cdot \frac{\sqrt{3}}{2}$ $5\sqrt{3}$ ft	The fishing net is hanging, dry, and clean. Nothing was wrapped in it.
9	If $\sin\theta = \frac{1}{2}$ for an acute angle θ, find θ.	**D.** 30°	$\theta = \sin^{-1}(\frac{1}{2})$ 30°	Mr. Tern was counting seagulls from the Spiral Stairs. His notebook has the times.
10	Find x to the nearest tenth: tan 35° = $\frac{x}{10}$	**C.** 7.0	$x = 10 \cdot \tan 35°$ $x \approx 7.0$	The barrels by the dock are sealed and heavy, so nothing floated away in them.

Wrong-Answer Map

The Lighthouse Lantern Mystery · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	A. $2\sqrt{7}$	subtracted squares	Every backpack at the ferry gate was unzipped and searched. Just lunches and rain jackets.
1	B. 100	forgot square root	Keeper Odell was sketching the sea view from the Cliff Path. A hiker snapped his photo.
1	C. 14	added legs	Dr. Gale was checking the wind gauge in the Lamp Room. The weather log is stamped with the hour.
2	A. 8	subtracted lengths	The backpacks on the hook rack were all checked. Only sweaters and sandwiches inside.
2	B. $\sqrt{194}$	added squares	Captain Marlow was tying his boots on the Cliff Path. A jogger waved to him.
2	C. 18	added leg and hypotenuse	Dr. Gale was climbing the Spiral Stairs with a barometer. Three tourists counted her steps.
3	B. $5\sqrt{3}$	used 30-60-90 ratio	Ms. Brine was selling mackerel from a cart on the Cliff Path. Her receipts have the time.
3	C. 10	added the legs	Dr. Gale was talking to the crew at the Harbor Dock about the storm. The dockmaster wrote it down.
3	D. 25	squared a leg only	The only backpack found was empty and flat. Nothing was carried in it.
4	A. $8\sqrt{2}$	multiplied by $\sqrt{2}$	Dr. Gale was buying a crab roll at the Fish Market. The vendor remembers her umbrella.
4	C. $4\sqrt{3}$	used 30-60-90 ratio	A backpack was weighed on the dock scale. It matched its owner's gear, nothing extra.
4	D. 4	leg equals half hypotenuse	Mr. Tern was watching puffins from the Cliff Path with binoculars. His bird list has the times.
5	B. 8	gave hypotenuse	Dr. Gale was borrowing a tarp in the Boathouse. Nico Reyes signed her out.
5	C. $2\sqrt{3}$	halved the answer	Search dogs sniffed every backpack at the gate. None of them reacted.
5	D. $4\sqrt{2}$	used 45-45-90 ratio	Nico Reyes was measuring the Cliff Path fence for a new railing. His tape notes are dated.
6	A. $\frac{3}{4}$	used tan A	Mr. Tern was sketching a gull nest on the Cliff Path until sunset. His sketchbook has the date.
6	C. $\frac{4}{5}$	used cos A	Dr. Gale was cleaning the lens in the Lamp Room with Keeper Odell. The brass is still shiny.
6	D. $\frac{5}{3}$	inverted ratio	The straps on every backpack were clean and unworn. None had carried anything heavy.
7	A. $\frac{4}{3}$	inverted tan	Dr. Gale was sending a radio report from the Harbor Dock. The signal log shows the time.
7	B. $\frac{4}{5}$	used cos A	Nobody left the lighthouse with a backpack that day. The gate guard kept a tally.
7	D. $\frac{3}{5}$	used sin A	Captain Marlow was waiting out a gust on the Cliff Path. A kite flyer saw him there.

8	A. 10√3 ft	misapplied 30-60-90 ratio	Keeper Odell was trimming weeds along the Cliff Path. The gardener's sheet backs him up.
8	B. 5√2 ft	used 45-45-90 ratio	A backpack was opened at the lookout. It held only a thermos and a map.
8	D. 5 ft	used cos instead of sin	Dr. Gale was resting on the Spiral Stairs with a cup of tea. A lighthouse guide sat with her.
9	A. 60°	used cos inverse	Dr. Gale was mending a rain gauge in the Boathouse. The boat builder's apprentices watched her.
9	B. 26.6°	treated ratio as angle	The lost-and-found backpack was checked twice. It held only socks and a flashlight.
9	C. 45°	guessed common angle	Ms. Brine was setting up a stall near the Cliff Path at noon. Her permit slip is stamped.
10	A. 8.2	used cos 35°	The backpack on the bench belonged to a tourist. It held only snacks and sunscreen.
10	B. 14.3	used 10 divided by tan	Dr. Gale was sampling chowder at the Fish Market. The stall owner kept her receipt.
10	D. 5.7	used sin 35°	Nico Reyes was carving a sign for the Cliff Path trailhead. Two hikers stopped to watch.